

42.510 Statistical Learning for Data Science

Matrix factorization II: PCA applied to Social Progress Index

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Week 9, Lecture 2

Outline

- Social Progress Imperative
- Recall: Matrix factorization
- Principal Component Analysis

Social Progress Imperative

Opinion

'We're No. 28! And Dropping!'

A measure of social progress finds that the quality of life has dropped in America over the last decade, even as it has risen almost everywhere else.



By Nicholas Kristof
Opinion Columnist

Sept. 9, 2020



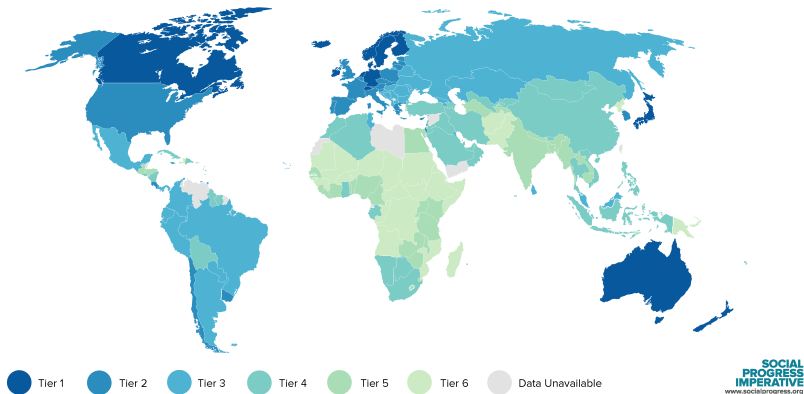
Graphic

Social Progress Imperative

- Economic progress/ growth perhaps does not provide a holistic view of development, prosperity or well-being of its citizens.
- A relevant question to ask is if a society/nation meets the basic needs of its citizens in terms of
 - ▶ healthcare
 - ▶ education
 - ▶ clean water
 - ▶ human rights
 - ▶ environmental quality
 - ▶ gender equality
 - ▶ public/ personal safety
 - ▶ ...
- Topic is of interest to social scientists, policy makers, governments, businesses, ...
- Note: Nobel Prize in Economics (2019): Experimental approach to alleviating global poverty
 - ▶ improving education (W. Kenya), preventive healthcare subsidies (India)
 - ▶ Per capita income/ GDP are NOT mentioned directly.

Social Progress Imperative: map

2020 Social Progress Index



Video

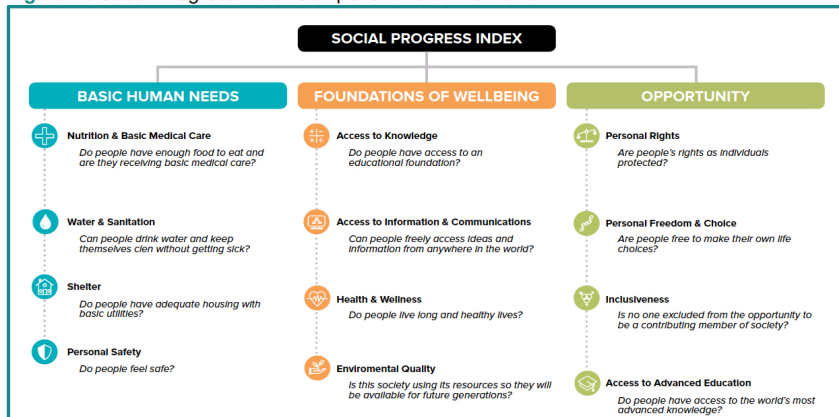
Social Progress Index

The Social Progress Index is a well-established measure, published since 2013, that is meant to catalyze improvement and drive action by presenting social outcome data in a useful and reliable way. Composed of multiple dimensions, the Social Progress Index can be used to benchmark success and provide a holistic, transparent, outcome-based measure of a country's well-being that is independent of economic indicators. Policymakers, businesses, and countries' citizens alike can use it to compare their country against others on different facets of social progress, allowing the identification of specific areas of strength or weakness.

- taken from the Methodology summary

Social Progress Index

Figure 1 / Social Progress Index Component-Level Framework



Social Progress Index

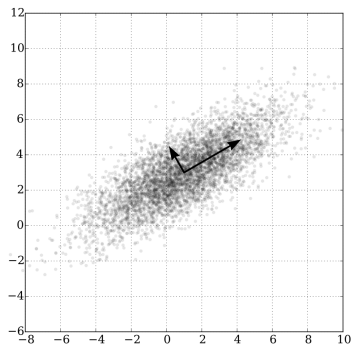
Figure 2 / Social Progress Index Indicator-Level Framework



Key questions

- ① What can we understand about social progress from the raw data?
- ② Can we identify where the major differences in social progress occurs?
- ③ Can we propose some kind of social progress index from our analysis.

Principal Component Analysis



- n = Number of observations
- p = Number of feature variables
- $X_1, \dots, X_p$ = feature variables

- $n \times p$ data matrix:

$$\mathbb{X} = \begin{bmatrix} x_{11} & \dots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}.$$

- Which directions shows most variability?
 - ▶ or, which hyperplane has shortest average distance to data?
- **Assume:** $\sum_{i=1}^n x_{ij} = 0, \quad \forall j = 1, \dots, p.$
- Compute sample covariances:

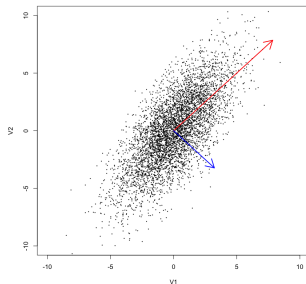
$$\widehat{\text{Cov}}(X_j, X_k) = \frac{1}{n-1} \sum_{i=1}^n x_{ij} x_{ik}, \quad j \neq k.$$

- Sample Covariance matrix:

$$\mathbf{S} = \frac{\mathbb{X}^\top \mathbb{X}}{n-1}.$$

Low dimensional representation of data

Generate 5000 data points from bivariate normal with zero mean, variances 5, 10 and covariance 5.



The red arrow show direction of maximum variability ($\lambda_1 v_1$) and blue arrow show second orthogonal direction of maximum variability ($\lambda_2 v_2$).

Low dimensional representation of data: PCA

- **Goal:** Minimize **MSE** s.t. $\|\mathbf{v}\| = 1$.
- Note that for a line/direction given by $\mathbf{v}$ with $\|\mathbf{v}\| = 1$, we have

$$\begin{aligned}\text{SSE} &= \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{v}^\top \mathbf{x}_i) \mathbf{v}\|^2 = \sum_{i=1}^n (\mathbf{x}_i^\top - (\mathbf{v}^\top \mathbf{x}_i) \mathbf{v}^\top) (\mathbf{x}_i - (\mathbf{v}^\top \mathbf{x}_i) \mathbf{v}) \\ &= \sum_{i=1}^n \mathbf{x}_i^\top \mathbf{x}_i - 2 \sum_{i=1}^n (\mathbf{v}^\top \mathbf{x}_i)^2 + \sum_{i=1}^n (\mathbf{v}^\top \mathbf{x}_i)^2 \\ &= \sum_{i=1}^n \sum_{j=1}^p x_{ij}^2 - \sum_{i=1}^n \mathbf{v}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{v} = \sum_{i=1}^n \sum_{j=1}^p x_{ij}^2 - (n-1) \mathbf{v}^\top \mathbf{S} \mathbf{v}.\end{aligned}$$

- Thus equivalent goal: Maximize $\mathbf{v}^\top \mathbf{S} \mathbf{v}$ s.t. $\|\mathbf{v}\| = 1$.
- From the previous class: optimal objective function value is the maximum eigenvalue of $\mathbf{S}$ given by λ_1 and the maximizer is its corresponding eigenvector $\mathbf{v}_1$.
- Thus $\mathbf{v}_1$ is the first principal component explaining λ_1 amount of variance (variability).

Low dimensional representation of data: PCA (contd...)

- Next step is to Maximize $\mathbf{v}^\top \mathbf{S} \mathbf{v}$ s.t. $\|\mathbf{v}\| = 1$ and $\mathbf{v} \perp \mathbf{v}_1$.
- and so on...
- Recall that we assumed that the data has mean 0 across all variables, i.e,

$$\sum_{i=1}^n x_{ij} = 0, \quad \forall j = 1, \dots, p.$$

- Hence total variability in the data is

$$\begin{aligned} \frac{1}{n-1} \sum_{i=1}^n \sum_{j=1}^p x_{ij}^2 &= \text{Trace} \left(\frac{\mathbf{X}^\top \mathbf{X}}{n-1} \right) = \text{Trace}(\mathbf{S}) \\ &= \text{Trace}(\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top) \quad (\text{eigendecomposition}) \\ &= \sum_{j=1}^p \lambda_j. \end{aligned}$$

- Hence choosing the first k components, the variability explained is $\sum_{j=1}^k \lambda_j$.

PCA: How many to choose?

- Customarily all variables are standardized to have variance 1.
- Hence, total variability in the data is

$$\sum_{i=1}^p S_i^2 = \sum_{j=1}^p \frac{1}{n-1} \sum_{i=1}^n x_{ij}^2 = p = \sum_{j=1}^p \lambda_j.$$

- One option is to take all principal components for which $\lambda_i > 1$, i.e., all the components which explain variability of more than 1 variable.
- Another option to use the **scree plot** of consecutive eigenvalues (variances) and look for an *elbow* in the plot.
- We may also fix the percentage of variability we want to explain to choose the number of PCs.

R implementation

- `prcomp`
- `princomp`

Limitations of PCA

- PCA depends on the scaling of the variables and hence one needs to scale the variables, essentially bringing them all down to the same scale.
- The eventual principal components loses some interpretability.
- We assume that a linear relationships between variables can explain the total variability.

Resources

- Friedman, J., Hastie, T., and Tibshirani, R. (2001). The Elements of Statistical Learning. Vol. 1. 10. Springer series in statistics, New York.
- <https://www.datacamp.com/community/tutorials/pca-analysis-r>
- <https://towardsdatascience.com/principal-component-analysis-pca-101-using-r-361f4c53a9ff>